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$$2. \rho_\nu d\nu = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{\exp(\frac{h\nu}{k_B T}) - 1} d\nu = -\rho_\lambda d\lambda$$

$$\lambda \nu = c \Rightarrow \lambda d\nu + \nu d\lambda = 0 \Leftrightarrow \frac{d\nu}{d\lambda} = -\frac{\nu}{\lambda} = -\frac{c}{\lambda^2}$$

$$\Rightarrow \rho_\lambda = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{\exp(\frac{h\nu}{k_B T}) - 1} \cdot \left(-\frac{c}{\lambda^2}\right)$$

$$= \frac{8\pi h}{\lambda^5} \cdot \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1} \cdot \frac{c}{\lambda^2} = \frac{8\pi hc}{\lambda^5} \cdot \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1}$$

$$\left. \frac{d\rho_\lambda}{d\lambda} \right|_{\lambda=\lambda_m} = 8\pi hc \left[-\frac{5}{\lambda^6} \cdot \frac{1}{\exp(\frac{hc}{\lambda k_B T}) - 1} + \frac{1}{\lambda^5} \cdot \left(-\frac{\exp(\frac{hc}{\lambda k_B T})}{(\exp(\frac{hc}{\lambda k_B T}) - 1)^2} \right) \cdot \left(-\frac{hc}{\lambda^2 k_B T} \right) \right]_{\lambda=\lambda_m} = 0$$

$$\Rightarrow -5\lambda_m + \frac{hc}{k_B T} \cdot \frac{\exp(\frac{hc}{\lambda_m k_B T})}{\exp(\frac{hc}{\lambda_m k_B T}) - 1} = 0$$

$$\Leftrightarrow 5\lambda_m T = \frac{hc}{k_B} \cdot \left(1 + \frac{1}{\exp(\frac{hc}{\lambda_m k_B T}) - 1} \right)$$

記 $x = \lambda_m T$.

$$5x = \frac{hc}{k_B} \cdot \left[1 + \frac{1}{\exp(\frac{hc}{k_B x}) - 1} \right]$$

此方程有唯一解 $x \approx 2.9 \times 10^{-3} \text{ m} \cdot \text{s} = b$

$$\text{即 } \lambda_m T = b \approx 2.9 \times 10^{-3} \text{ m} \cdot \text{s}$$

$$3. \lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = 0.70892 \text{ nm}$$

$$5. E = \frac{p^2}{2m} + \frac{1}{2} k q^2 \Leftrightarrow \frac{p^2}{\frac{2E}{k}} + \frac{p^2}{2mE} = 1$$

$$\oint p dq = \pi \cdot \sqrt{\frac{2E}{k}} \cdot \sqrt{2mE} = 2\pi E \sqrt{\frac{m}{k}} = (n + \frac{1}{2})h$$

$$\Rightarrow E = \frac{(n + \frac{1}{2})h}{2\pi} \sqrt{\frac{k}{m}}$$

$$6. h \cdot \frac{c}{\lambda} \geq m_0 c^2 \Rightarrow \lambda \leq \frac{h}{m_0 c} = 2.4271 \times 10^{-12} \text{ m}$$

$$7. E_k = \frac{3}{2} k_B T = 1.5 \times 1.38 \times 10^{-23} \approx 2.1 \times 10^{-23} \text{ J}$$

$$E_0 = m_0 c^2 = \frac{0.91 \times 10^{-31}}{1.67 \times 10^{-27}} \times (3.0 \times 10^8)^2 = 15.03 \times 10^{-14} \text{ J} \gg E_k$$

$$\text{因此 } p \approx \sqrt{2mE_k}$$

$$\lambda = \frac{h}{\sqrt{2m_{He} E_k}} = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 4 \times 2.1 \times 10^{-23}}} \approx 1.26 \times 10^{-9} \text{ m} = 1.26 \text{ nm}$$

思維導圖

普朗克公式：輻射能量在吸收或發射時以整個量子進行，頻率為 ν 的輻射粒子的能量 $E = h\nu$ ($h = 6.626 \times 10^{-34} \text{ J s}$)

$$\text{Planck - Einstein 光量子論: } P_\nu = \frac{8\pi h \nu^3}{c^3} \cdot \frac{1}{\exp(\frac{h\nu}{k_B T}) - 1} \quad K_e = h\nu - E_0$$

$$\text{Compton 效應: } \Delta\lambda = \frac{4\pi h}{m_0 c} \sin^2 \frac{\theta}{2} = \frac{h}{m_0 c} (1 - \cos \theta)$$

$$\text{Bohr 原子結構: 氫原子能量 } E_n = -R_H h c \cdot \frac{1}{n^2} \quad R_H = 1.097 \times 10^7 \text{ m}^{-1}$$

定態假設：原子在「穩定地」處於一系列能量態 $\{E_n\}$ 中。

此系列能量稱為定態

躍遷假設：在 E_n 和 E_m 定態中躍遷時，

吸收或發射光輻射的頻率

$$\nu = \frac{|E_n - E_m|}{h}$$

量子化條件：電子軌道角動量只能是 h 的整數倍

$$[\text{Sommerfeld: } \oint p dq = (n + \frac{1}{2})h]$$

De Broglie 假說：物質波 $E = h\nu = \hbar\omega$
 $p = \hbar k$